SMART TIME TABLE GENERATOR: A Full Ecosystem for Academic Scheduling

Streamlined, Automated, and Human-Centric.

Made with GAMMA

The Chaos of Manual Scheduling

Manual scheduling is broken—complexity scales fast: thousands of variables, hundreds of thousands of potential conflicts. Current pain points create operational risk and staff burnout.

Teacher Conflicts

Multiple assignments, overlapping time slots, unresolvable availability.

Room Overload

Insufficient capacity mapping and last-minute venue clashes.

Invalid Paths

Sequencing errors for modules and pre-requisite constraints.

Wasted Hours

Manual edits and reconciliation consume administrative time.

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Head of Department Dashboard (React / TypeScript / MUI)

Centralized admin interface designed for speed, clarity, and secure operations. Built with modern web stack and UX patterns for power users.

Automated Synthesis

Backtracking algorithm resolves constraints and minimizes conflicts at scale.

Drag & Drop

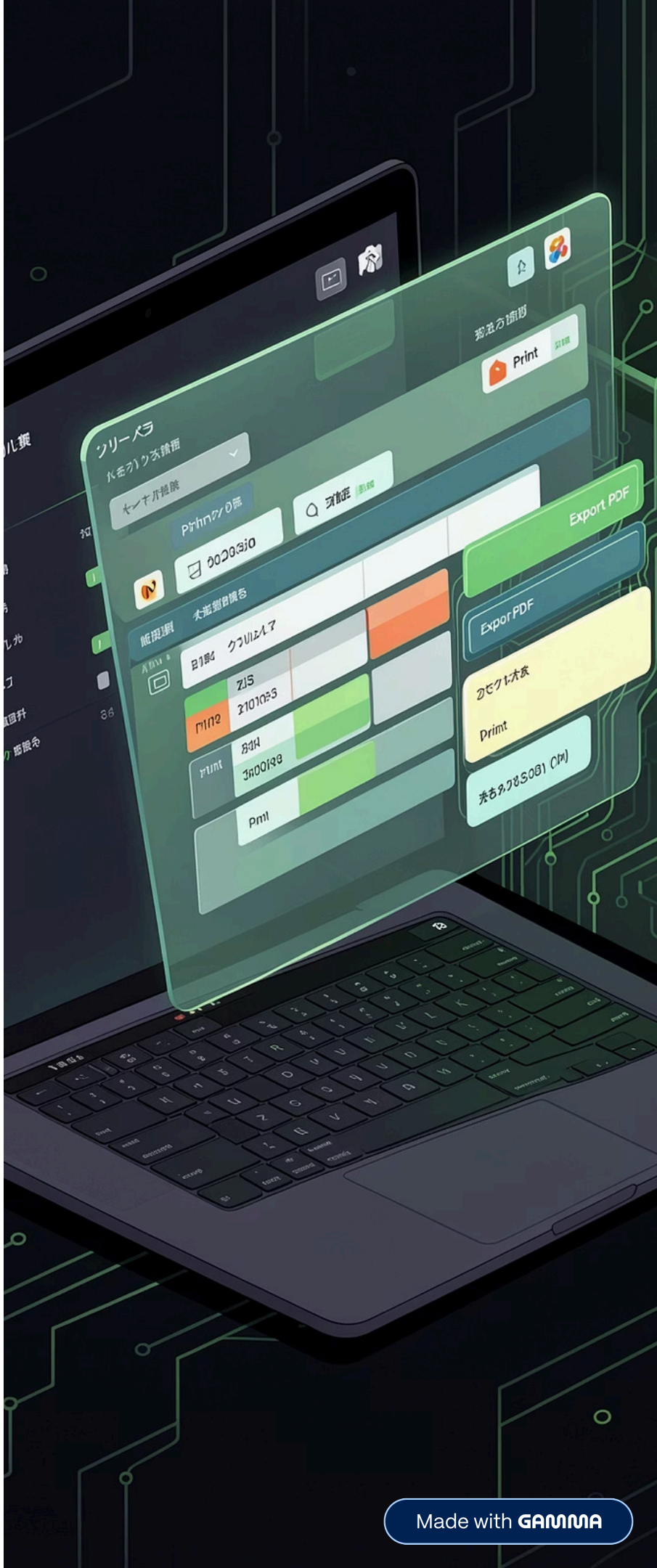
Post-generation edits via intuitive drag-and-drop for rapid manual adjustments.

Export

Export schedules to PDF and CSV for reporting and archival workflows.

Auth Guard

Secure access using cookie-based auth guard and role-aware permissions.



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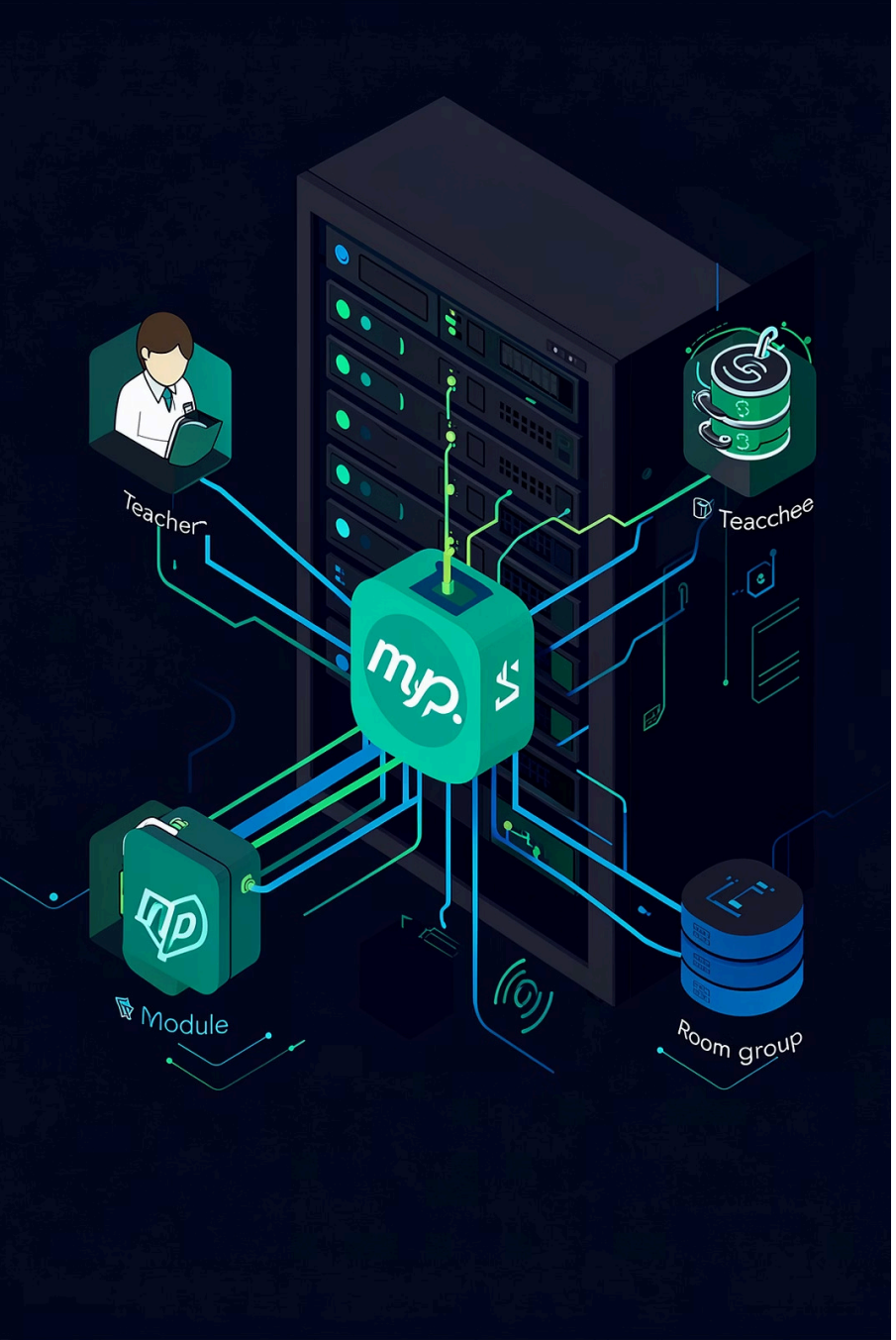

Scalable Backend & Data Integrity

Robust architecture focused on consistency, auditability, and horizontal scale. Designed for predictable, repeatable scheduling operations.

Technology Stack

- Node.js & Express — stateless API layer
- MySQL — ACID-compliant relational store
- RESTful endpoints with strong validation

Structured Data Model

Normalized entities: Teachers, Modules, Rooms, Groups. Referential integrity and versioned edits ensure reliable reconciliation.

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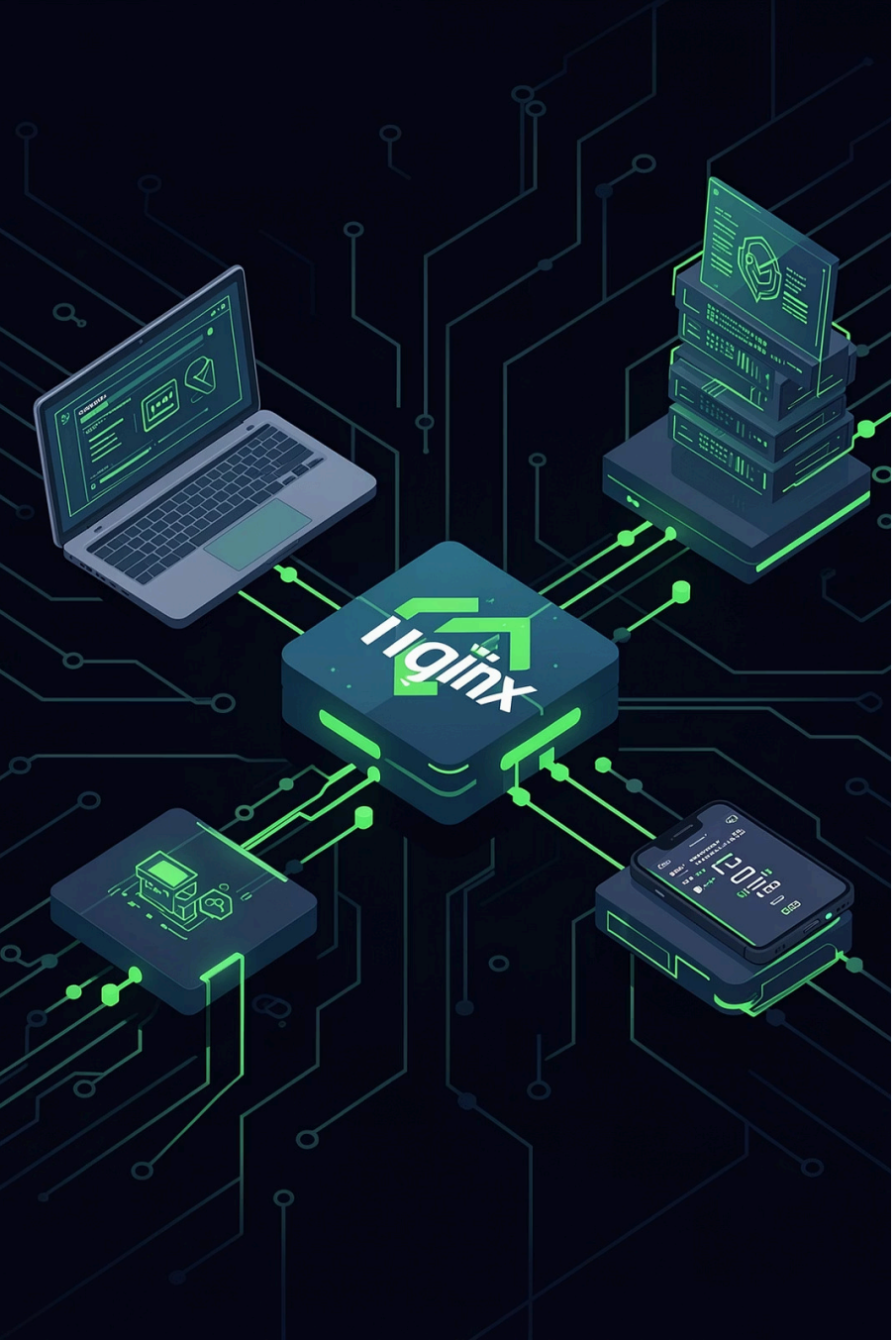

Robust Infrastructure & Hosting

Operational reliability and seamless connectivity across clients and services. Designed for maintainability and low-latency response.

Nginx Reverse Proxy

Central entry point for TLS termination, caching, and load balancing.

Service Segmentation

Separate Web, API, and Worker layers for resilience and independent scaling.

CI / CD

Automated pipelines for builds, tests, and safe deployments to staging and production.

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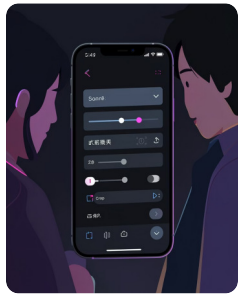
Cross-Platform Access (Flutter Mobile App)

Native-feel experiences for students and teachers—fast, offline-capable, and ready for integration.

Student View

Personalized timetables, push notifications for changes, quick access to room maps and instructor info.



Teacher View

View schedules, submit edit requests, and approve proposed changes—mock data ready for API integration.

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Feature: Offline Mode & Synchronization

Continuous availability and robust merge strategies protect data when connectivity is intermittent. Designed to minimize conflicts and preserve admin intent.

Local Caching

Read and browse timetables offline with immutable snapshots.

Offline Edits

Admins and teachers can stage changes while offline; changes are versioned locally.

Smart Sync

Conflict-aware merge on reconnect with human review for complex edits.

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Project Management & Scalability

Development workflows and priorities to reach reliable, maintainable releases—aligned to departmental needs and scale targets.

Source Control

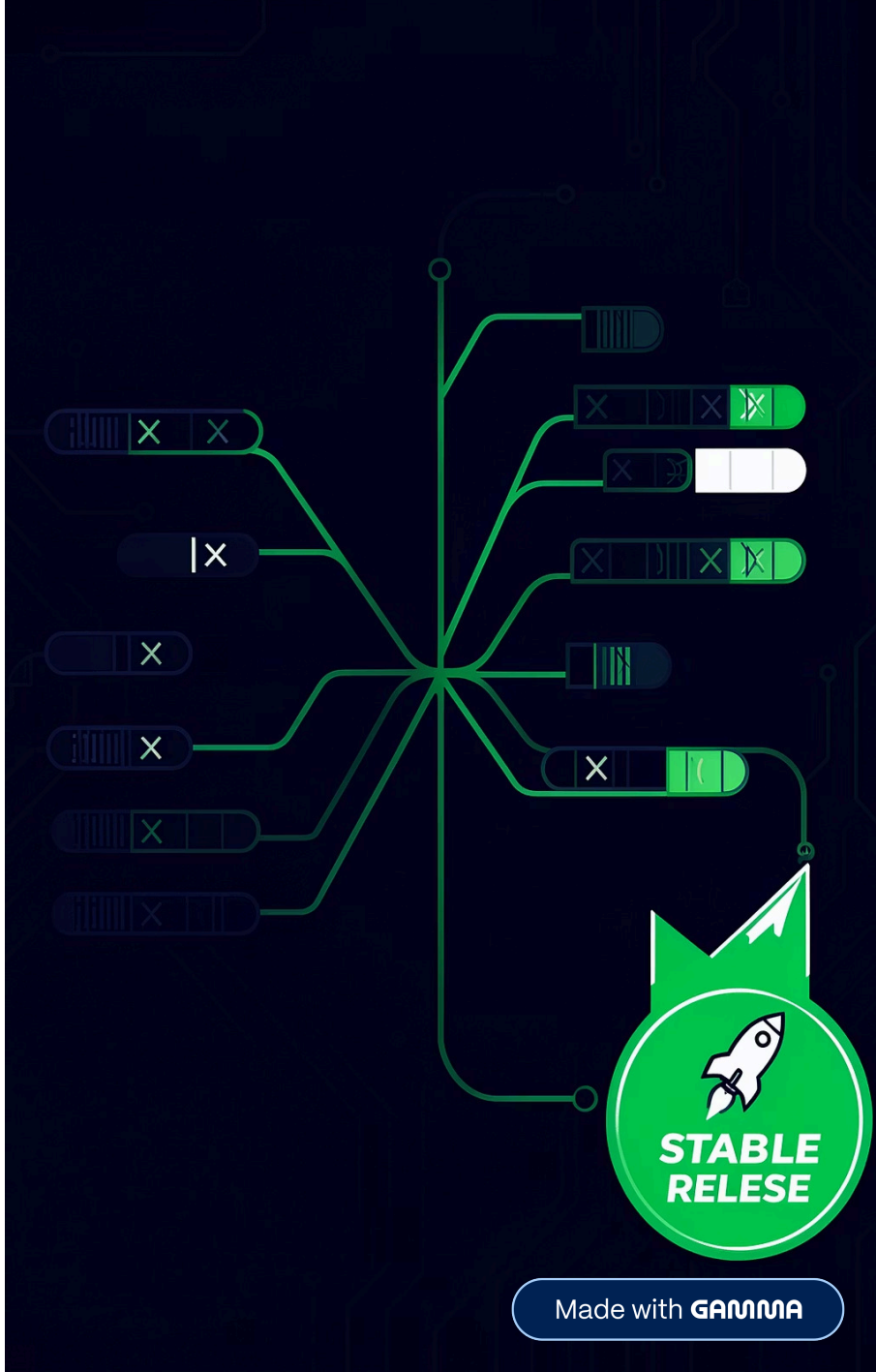
Git + GitHub for versioning, protected branches, PR reviews, and CI checks to ensure stability.

- Feature branches per module
- Signed releases and changelogs

Next Steps

1. Full Web / Mobile API integration and contract testing
2. Load testing and scaling to multi-campus deployments
3. Pilot with 1–2 departments, then broader roll-out

Contact: Technical lead for demo, architecture review, and pilot planning.



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